

METHODS AND TECHNIQUES

WorldCuP: A database of checklists documenting the World's Cultivated Plants and their distributions

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Societal Impact Statement

Human life would be unrecognizable without the benefits of cultivated plants, including crops, ornamentals, or plants with other uses. Despite their importance in daily life and prevalence across landscapes, we know surprisingly little about global cultivated plant distributions. To close this gap, we digitized checklists listing vascular plants that were selected, planted, and cared for by people in regions worldwide. We harmonized and compiled this data to form the World Cultivated Plants (WorldCuP) database. The WorldCuP database, which links plant species to the regions in which they are cultivated, enables new research in ecological impacts, conservation, ethnobotany, and biological invasions.

Summary

- People have relied on cultivated plants for thousands of years in order to meet basic needs and to provide diverse sociocultural and environmental services. The ubiquity of cultivated plants in human-dominated landscapes has important implications for the structure and functioning of communities, ecosystems, and regional species pools. A thorough documentation of the global distributions of cultivated plant species is necessary to stimulate research on the impacts of cultivated plants and to support large-scale integration of historical and cultural knowledge about cultivated plants.
- Here, we present the World Cultivated Plants (WorldCuP) database: a publicly available living database of 3050 checklists of cultivated vascular plant taxa in 2718 globally distributed regions spanning 156 countries. The database has accompanying taxonomic information (original and standardized names), geographic data (simple features for each region), and references for original sources. Data were gathered from 597 printed and online sources including regional floras, botanical garden collections, nursery offerings, and surveys of domestic gardens.
- Although most sources were modern (70.6% published after 2000), historical sources dating from 1375 are included. The WorldCuP database includes 102,243

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accepted cultivated plant taxa, including 91,358 species, 6064 subspecies, 4722 varieties, and 99 forms. WorldCuP has nearly global coverage, however, there are notable absences in parts of Africa, southwest and central Asia, and a number of islands.

- WorldCuP is the most comprehensive collection of cultivated plant species occurrences to date and will be a new resource for regional studies or global analyses of cultivated plants.

KEYWORDS

crops, cultivation, garden plants, global distributions, ornamental plants, plant diversity

1 | INTRODUCTION

While nearly all animals depend on plants, humans have a special relationship with them. We cultivate plants, selecting individuals to be planted, tended to, and propagated. Cultivated plants are important for meeting our basic needs, providing food, shelter, fuel, and medicine. Furthermore, cultivated plants enrich our cultures, delivering aesthetic and spiritual benefits, as well as the environments we inhabit, improving soil quality, regulating water/climate, and controlling erosion (Martín-López et al., 2018).

The history of plant cultivation dates many thousands of years. In the late Pleistocene, people directly and indirectly selected for useful plants by acting as ecosystem engineers, burning vegetation and hunting megafauna (Allaby et al., 2022; Boivin et al., 2016). Agriculture and horticulture (i.e., plant cultivation) began independently in at least 11 globally widespread regions between 12,000 and 4000 years ago (Larson et al., 2014). Gardens, delineated spaces where people cultivate plants, are known from archaeological and historical evidence across disparate regions and cultures, including Early Bronze Age Mesopotamia (c. 3400 BCE, Farrar, 2016), Old Kingdom of Egypt (c. 2613–2494 BCE, Daines, 2008), Shang Dynasty China (c. 1600–1050 BCE, Pearson, 2023), Archaic Greece (c. 500 BCE, Farrar, 2016), and Classic Maya (660 CE, Farahani et al., 2017). Increasing numbers of plant species were brought into cultivation during the era of European colonization and imperialism from the late 15th to the mid-20th century. During this period, cultivated plants were exchanged across a global network of botanical gardens, commercial nurseries, research stations, and governmental institutions for their potential political and economic benefits (Brockway, 1979; Drayton, 2000; McCracken, 1997). From the late-20th century onward, as globalization has intensified long-distance trade networks and the global human population has rapidly increased, the rates of cultivated plant production and exchange have correspondingly increased (Dehnen-Schmutz et al., 2010; Seebens et al., 2018).

Cultivated plants form the backdrop of human-inhabited environments, which make up approximately 30% of Earth's land area (Ellis & Ramankutty, 2008). Cultivated plants are integral components of such environments, affecting other plants, pollinators, seed dispersers, herbivores, soil biota, and the abiotic environment (Altieri, 1999). Among

these effects, the cultivation of crops and garden plants has intensified global nitrogen and phosphorous cycling (Liu et al., 2010; MacDonald et al., 2011) and reduced insect diversity (Outhwaite et al., 2022). It has also increased local plant diversity in urban areas (Loram et al., 2008; Seitz et al., 2022) and shifted the functional composition of pollinator communities (Baldock et al., 2015; Udy et al., 2020). In addition, thousands of non-native plant species have been introduced to new regions for cultivation, and many cultivated plants can persist in their new habitats without the aid of humans (Hulme et al., 2018; van Kleunen et al., 2018). Cultivation, which involves propagating and distributing non-native plant species widely throughout the landscape, is an important pathway for invasion (Hulme et al., 2008; Reichard & White, 2001), and invasive plants have well-recognized negative impacts on native communities and ecosystems (Bradley et al., 2019; Thakur et al., 2025; Vilà et al., 2011).

A better understanding of the diversity and distributions of cultivated plants will foster diverse research pathways. First, knowledge about the historical and present-day cultivated flora is valuable from a sociocultural perspective (Prescott-Allen & Prescott-Allen, 1990; Schiebinger & Swan, 2005). Next, cultivation plays an important, but under-investigated, role in the ex situ conservation of threatened plant species (Ismail et al., 2021; Mounce et al., 2017). Finally, understanding which species are cultivated in particular regions is needed to predict the effects of cultivated plants on communities, ecosystems, and regional species pools worldwide (Castro-Díez et al., 2019; Pearse et al., 2018). Increasing digitization of material publications has made records of cultivated plants, once widely scattered in books, botanic garden records, and nursery catalogs, more accessible. Despite the accumulation of accessible cultivated plant records, global-scale data linking plant species to the regions in which they are cultivated are still not available.

Several previous efforts have aimed to catalog the world's cultivated plants. In order to record plant resources that have been used across the world for their potential to support food and resource security, Rudolf Mansfeld compiled an encyclopedia listing agricultural and horticultural crop species (Mansfeld, 1959), which was later published online in 1998 by the Leibniz Institute of Plant Genetics and Crop Plant Research (IPK), Gatersleben. Mansfeld's database excludes certain categories of cultivated plants, namely, ornamentals and

forestry plants, and does not systematically describe the regions where each taxon is cultivated. Related global datasets cataloging the native ranges of crop species (Zeven & De Wet, 1982) and the wild progenitors of crop species (Dempewolf et al., 2014; Milla, 2020) have also been published. There have also been efforts to catalog plants with documented human uses. Most notably, the Royal Botanic Gardens, Kew compiled the World Checklist of Useful Plant Species (WCUPS) from a set of 13 large datasets of useful plants (Diazgranados et al., 2020). Similarly in this database, the regions where each taxon is used are not listed. While useful plants overlap with cultivated plants, useful plants can be harvested from the wild and thus are not necessarily intentionally cultivated by people. Finally, taxa cultivated in botanic gardens across the world have been cataloged in the Botanic Gardens Conservation International (BGCI) Plant-Search database (http://www.bgci.org/plant_search.php); however, the data describing taxa cultivated in each garden are not publicly available. Collectively, currently existing databases provide information about subsets of cultivated plant diversity without directly linking plant species to the specific regions where they are cultivated. Here, we address this absence by presenting the World Cultivated Plants (WorldCuP) database, a publicly available database of cultivated vascular plant taxa in globally distributed regions.

2 | MATERIALS AND METHODS

2.1 | Data acquisition

Lists in WorldCuP came from any source listing cultivated species that were planted and maintained in a particular region or location, including plants cultivated as ornamentals, crops, for medicinal use, as fodder/forage, or for other uses. Plants that are useful but not cultivated and spontaneous plants in human-modified habitats were excluded from the database. Sources included floras or checklists of cultivated plants, living collections from botanic gardens or arboreta (or, if living collection data were not available, from *indices seminum*, i.e., lists of seeds from plants cultivated at one botanic garden that are on offer to other botanic gardens), lists of taxa for sale in nurseries or seed houses, governmental and nongovernmental reports and databases, and surveys of domestic gardens or farms (including homegardens, lawns, or street plantings). Sources were published as books, academic journal articles, online databases, websites, or as gray literature (e.g., reports published by governmental agencies or nongovernmental organizations).

Sources were identified over a 5-year period using internet searches for “cultivated plants OR flora of,” “list of plant species cultivated in,” “garden plants OR flora of,” and “ornamental plants in,” followed by the names of a country or region, with search terms translated into the language(s) of that region using online translation tools (e.g., Google Translate). Beginning in December 2021, sources were also identified using a weekly Web of Science Alert: “(urban OR homegarden OR agroforest*) AND (cultivat* OR horticultur* OR ornamental) AND (diversity OR composition OR floristic).” Version 1.0 of

the WorldCuP database includes sources that were identified before August 28, 2025.

Sources in any language were included as long as we could determine that they met requirements using online translation tools when necessary. Historical sources were also included; however, we only included sources from before the adoption of binomial nomenclature if taxon names were converted to modern-day scientific names by garden historians (note that only 0.23% of sources predate 1760). We only included taxa that were defined to the species or infraspecies level. We included cultivars as long as they had a specific epithet (e.g., we included *Berberis thunbergii* DC. ‘Silver Beauty’ but not *Rosa* ‘La France’).

WorldCuP is a living database, meaning that sources are continually identified, digitized, cleaned, and added to the database with each new version. A complete list of included sources, including full citations in the generic BibTeX format, is provided within the WorldCuP database (the “reference” table), and a complete bibliography of the sources in Version 1.0 is provided in the [Supporting Information](#). Original authors of the data sources are not co-authors of the WorldCuP database but are credited via citation. Thus, users of the WorldCuP database should cite both the present paper and the original data sources used.

2.2 | Taxonomic harmonization

Taxonomic standardization and insertion into the WorldCuP database were carried out per taxon list. Some sources included multiple taxon lists describing occurrences in different regions, and these sources were split into individual lists. We cleaned digitized taxon lists using the “PlantTaxonNameCleaner” Python program (<https://github.com/nlkinlock/PlantTaxonNameCleaner>), which standardizes spacing and capitalization, corrects genera misspellings, and replaces authorities with standard forms (Brummitt & Powell, 1992), in order to improve harmonization and to reduce the chance of getting a false match during taxonomic standardization.

We standardized cleaned taxon names using version 12 of the World Checklist of Vascular Plants (WCVP) taxonomic backbone (Govaerts et al., 2021). Specifically, we matched taxon names in WorldCuP to the WCVP taxonomic backbone and linked matched taxon names with the accepted scientific name for that taxon according to the synonymy in WCVP. We used the “wcvp_match_names” function in the “rWCVP” R package to match taxon names to the backbone, including authorities when present, which performs exact and fuzzy matching (Brown et al., 2023). For taxon names with multiple matches to the WCVP backbone, we selected the taxon name and the authorities with the highest Levenshtein similarity. Levenshtein distance is the minimum number of single character edits between two strings, and we converted this to a similarity metric using the “stringsim” function in the “stringdist” R package (van der Loo, 2014). We removed all matches for taxon names with multiple matches that could not be resolved using the previous methods, unless they were synonyms for the same accepted taxon name. For infraspecific taxon

names with no suitable match, we removed the infraspecific epithet and retained the standardized binomial name. We manually checked all fuzzy matches for accuracy and removed incorrect matches. For all taxon names with incorrect matches or with no match, we searched for synonyms in the Global Biodiversity Information Facility (<https://www.gbif.org>), World Flora Online (<https://www.worldfloraonline.org>), and World Plants (<https://www.worldplants.de>) databases. Cultivar names and unnamed hybrids were not taxonomically standardized, but they were included as original taxon names in WorldCuP. Note that accepted names in WCVP that are categorized as artificial hybrids are included in WorldCuP (e.g., *Musa* × *paradisica* L.). If desired, users can use a different taxonomic backbone to standardize original taxon names.

2.3 | Geographic data curation

Region information came from the sources. Sources provided cultivated plant species lists at specific locations, localities (villages, cities, or districts), states/provinces, countries, or other regions. The large majority of list regions used political rather than biogeographic boundaries. When provided, geographic boundary data came from the sources themselves. Otherwise, we extracted geographic boundary data for list regions that were administrative areas (e.g., countries, states/provinces, and districts) in the Global Administrative Areas Database (GADM) version 3.6 using shapefiles downloaded from <https://gadm.org>. Because the resolution of GADM is very high, GADM polygons were simplified using the Visvalingam method (which maintains shared boundaries) using the “ms_simplify” function in the “rmapshaper” R package (10% of points were retained and the Visvalingam simplification was weighted by 0.7, Teucher & Russell, 2023). We also extracted geographic boundary data for target regions defined using the World Geographic Scheme for Recording Plant Distributions from the Taxonomic Databases Working Group (TDWG, Brummitt, 2001) using shapefiles downloaded from <https://github.com/tdwg/wgsrpd>. For target regions not defined in GADM or TDWG, we extracted geographic boundary data from OpenStreetMap (<https://www.openstreetmap.org>). We extracted latitude and longitude coordinates for regions describing specific locations (e.g., botanic gardens, arboreta, parks, or nurseries), midpoints of localities for which there is no available geographic boundary data, or midpoints of multiple non-separable points (i.e., cultivated plants were assessed at multiple locations but the data were pooled). For the latter case, if coordinates were available for all locations, then we calculated the convex hull surrounding all points using the “st_convex_hull” function in the “sf” R package (Pebesma, 2018) and used the resulting polygon as the geometry. Other coordinates were extracted using Google Maps. All geometries used the World Geodetic System 1984 (WGS84) coordinate reference system.

Cleaning, taxonomic standardization, and geographic data curation were carried out using version 3 of the Python programming language and version 4 of the R programming language (R Core Team, 2025).

2.4 | Database structure and analysis

The WorldCuP database is a relational database, which maximizes data integrity while minimizing storage requirements, compiled using MySQL version 8.0. The WorldCuP database includes seven relational data tables with information about sources (“source” and “reference” tables), lists (“list” table), regions (“region” and “geom” tables), and taxa (“taxon_orig” and “taxon_acc” tables) (Table 1). The foundational table in WorldCuP is the “occurrence” linking table that links lists with taxa. Two additional linking tables link WorldCuP regions with TDWG regions (“region_tdwg”) and International Organization for Standardization (ISO) 3166-1 country codes (“region_country”). ISO country codes are derived from United Nations countries, dependent territories, and special areas of geographical interest. These are important when converting the data to a large spatial grain. WorldCuP also includes five validation tables that list all valid entries for categorical data in other tables. Metadata describing the columns in all WorldCuP tables are provided as comma-separated values (CSV) files along with the database.

We compared the WorldCuP database with other databases partially overlapping with cultivated plants by creating area-proportional Euler diagrams using the “euler” R package (Larsson & Gustafsson, 2018). This included (1) Mansfeld's World Database of Agricultural and Horticultural Crops (<https://mansfeld.ipk-gatersleben.de>) for agricultural and horticultural crop species, and (2) the World Checklist of Useful Plant Species (WCUPS, Diazgranados et al., 2020) for useful plant species that are not necessarily cultivated.

We estimated species accumulation curves for unique cultivated taxa in WorldCuP lists using 1000 random list permutations, calculating the mean and quantile-based 95% confidence intervals (CIs) of cumulative species richness across permutations for the database as a whole and separately within each global region (Europe, the Americas, Africa, Asia, and Oceania). For each permutation, we fitted Clench's equation (a reparameterized Michaelis-Menten equation, Clench, 1979; Soberón & Llorente, 1993), $S_{\text{obs}} = a \cdot I / (1 + b \cdot I)$, where S_{obs} is cumulative species richness and I is the cumulative number of lists, using nonlinear least squares via the “minpack.lm” R package (Elzhov et al., 2023). We calculated the asymptote of Clench's equation (a/b), which is an estimate of maximum species richness (S_{max}), and calculated the mean and 95% CIs of S_{max} and the proportion of S_{max} that was observed ($S_{\text{obs}}/S_{\text{max}}$) across permutations.

To explore the geographic coverage of the WorldCuP database, we resolved cultivated plant taxon occurrences to the country-level spatial grain by pooling WorldCuP lists within ISO countries and calculating taxon richness as the number of unique cultivated taxa per country (excluding lists with spatial extents larger than ISO countries). We also resolved occurrences to TDWG level-3 regions using the same methods. We estimated rarefied taxon richness by subsampling lists within each ISO country or TDWG level-3 region to a comparable sample size ($n = 5$, removing regions with fewer than n lists) and calculating the mean number of unique cultivated taxa per region after 1000 iterations.

To visualize the composition of cultivated plant taxa in ISO countries and in TDWG level-3 regions, we conducted a nonmetric

TABLE 1 Data, linking, and validation tables in the World Cultivated Plants (WorldCuP) relational database.

Name	Description
<i>Data tables</i>	
source	Lists unique sources included in WorldCuP, and also includes information about spatial extent, taxonomic subset, whether the source explicitly describes cultivated plants or excludes native plants, and whether the source is historical
reference	Lists unique references in WorldCuP. Includes information about publication year, title, and reference type, as well as complete BibTeX citations. Some sources are compiled from multiple references. Links references to 'source' table
list	Lists unique taxon lists in WorldCuP. Includes information about the year the list describes and whether the list includes data about native region, life form, or usage. A given source can include multiple lists, and each list describes only one region. Links lists to 'source' and 'region' tables
region	Lists unique regions in WorldCuP. Provides information about list regions, including whether regions are equivalent to TDWG regions or ISO countries, geometry type (polygon, midpoint, point), and longitude/latitude for (mid)points. Each list links with one region. Links regions to 'geom' table
geom	Stores spatial geometries (points or polygons) of list regions. Lists, regions, and geometries have one-to-one relationships
taxon_orig	Lists unique original taxon names (i.e., 'taxon_author_cv_orig,' including binomial names, authorities, and cultivars) in lists and the matched and standardized taxon names according to the WCVF taxonomic backbone. Matched names are the names in WCVF that match with original names and standardized names are the associated accepted names according to WCVF synonymy. Also provides information about taxonomic status, additional remarks about the standardization process, and whether an external database was used to identify synonyms. Links original taxon names to 'taxon_acc' table
taxon_acc	Lists unique WCVF accepted taxon names and authorities (i.e., 'taxon_author_acc,' including binomial names and authorities) in lists, and also includes associated WCVF IDs, family, taxonomic status, and WCVF data on geographic area, life form, and climate. Note that WCVF synonymy links a small number of taxa to an unplaced name or local biotype rather than an accepted name (or an artificial hybrid); these can be filtered out using the 'taxon status acc' column.
<i>Linking tables</i>	
occurrence	Each row describes a taxon occurrence in a list region. This is the foundational table of the WorldCuP database. Links 'taxon_orig' and 'list' tables
region_tdwg	Links list regions with TDWG regions (n.b., a given list region may link to multiple TDWG regions). Links 'region' and 'tdwg' tables
region_country	Links list regions with ISO countries (n.b., a given list region may link to multiple ISO countries). Links 'region' and 'country' tables
<i>Validation tables</i>	
tdwg	Lists all TDWG codes, names, and levels
country	Lists all three-letter ISO country codes (ISO 3166-1 alpha-3 codes) and their respective country names
geom_type	Lists types of geometries for list regions
ref_type	Lists reference type categories
extent_type	Lists categories describing the geographic extent of sources

multidimensional scaling (NMDS) ordination of Sørensen dissimilarity matrices using the "metaMDS" function in the "vegan" R package (with $k = 3$ dimensions and 2000 random starts, Oksanen et al., 2025).

3 | RESULTS AND DISCUSSION

Version 1.0 of the WorldCuP database includes 1,444,325 cultivated vascular plant taxon occurrences from 3050 lists across 2718 unique regions spanning 156 countries (Figure 1). Geographic coverage in WorldCuP is extensive except for parts of western and central Africa, southwest Asia, and Oceania. Coverage is also temporally extensive; lists span the time period from 1375 to 2025, though most lists were published after 2000 (70.6%, Figure 2c). Though historical lists were concentrated in Europe and North America, they were present in all global regions (Figure S1). Our broad coverage was achieved by collating 597 diverse printed and online sources (Figure 2b), 30% of

which were in languages other than English. Ignoring non-English publications has been previously identified as a source of bias in studies incorporating data from multiple sources (Amano et al., 2021; Konno et al., 2020). To our knowledge, WorldCuP is the most comprehensive publicly available collection of cultivated plant taxon occurrences available.

WorldCuP includes 436,983 vascular plant taxa; 57.1% of taxon names were accepted (including 5,793 artificial hybrids), 33.8% were synonyms (i.e., linked to an accepted name), and 9.1% were unplaced or had no match in the WCVF taxonomic backbone (Figure 2a). The 102,243 unique accepted taxa, including 91,358 species, 6064 sub-species, 4722 varieties, and 99 forms, came from 427 families. While some cultivated taxa had hundreds of occurrences in WorldCuP lists, most taxa had very few occurrences (Figure 2d).

WorldCuP includes most of the crop taxa in Mansfeld's database (88.7%), and the majority (66.6%) of the useful plants in the WCUPS database (Figure 3a). Because crops are by definition cultivated, taxa that are present in Mansfeld's database and missing in WorldCuP

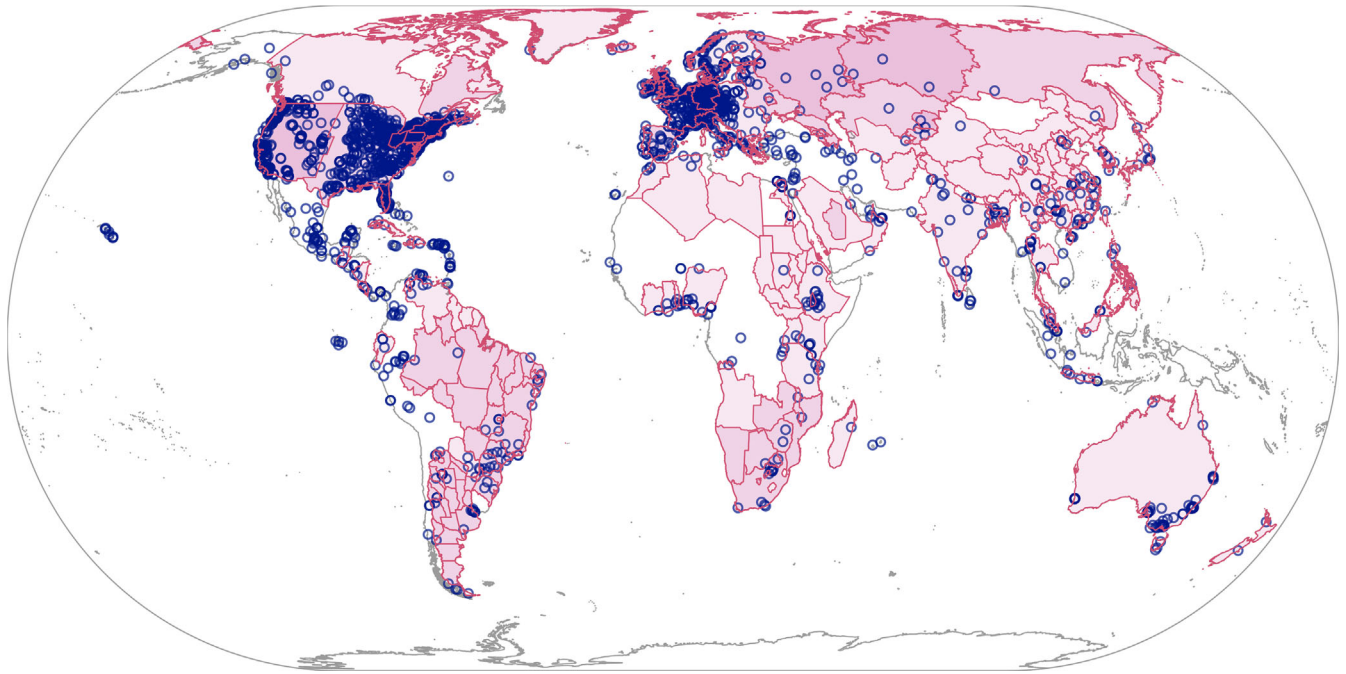


FIGURE 1 Map of regions in the World Cultivated Plants (WorldCuP) database. Point regions (in dark blue) represent locations (e.g., botanic gardens, parks, nurseries, and villages) and polygon regions (in pink) represent larger spatial grains (e.g., countries, regions, and cities). Polygons are translucent and can overlap, meaning that darker areas include multiple regions. The Eckert IV projection is used.

indicate gaps in WorldCuP's taxonomic coverage. In contrast, because some useful plants are not cultivated (e.g., wild-gathered useful plants), lower overlap with the WCUPS database reflects both WCUPS's inclusion of useful taxa that are not cultivated and taxonomic coverage gaps in WorldCuP. It is not possible to distinguish differences in inclusion criteria from actual gaps in taxonomic coverage for WCUPS versus WorldCuP, though the very high overlap of WorldCuP with Mansfeld's crop database suggests that a considerable portion of taxa in WCUPS but not WorldCuP are simply not cultivated. Most of the taxa in WorldCuP are not included in either Mansfeld's or the WCUPS database (72.9%, Figure 3a). Thus, WorldCuP overlaps with these major databases describing subsets of cultivated plants, but also includes many additional cultivated taxa.

Although crops and useful plants made up a relatively small proportion of the cultivated taxa in WorldCuP (6% crops, 26.2% useful plants, Figure 3a), they made up the majority of the 1,167,269 unique occurrences of accepted taxa in WorldCuP lists (28% crops from Mansfeld, 65% useful plants from WCUPS, Figure 3b). This means that commonly cultivated taxa tend to be crops or other useful plants. Indeed, the five most frequent taxa in WorldCuP lists were trees with primarily ornamental uses: *Ginkgo biloba* L., *Thuja occidentalis* L., *Gleditsia triacanthos* L., *Morus alba* L. (used in silk production), and *Acer platanoides* L. The five taxa with the broadest cultivated ranges (i.e., cultivated in the most countries) were primarily crops: *Psidium guajava* L., *Punica granatum* L., *Carica papaya* L., *Mangifera indica* L., and *Melia azedarach* L. (the latter is an ornamental tree with diverse medicinal and economic uses). On the other hand, rarely cultivated taxa, which make up the majority of cultivated taxon richness, are

likely cultivated as curiosities or for their scientific or conservation value (Brockway, 1979; McCracken, 1997; Mounce et al., 2017).

Data completeness of WorldCuP is difficult to assess. Garden owners, landscapers, and farmers can cultivate any desired taxon as long as its propagules are accessible. Regional inventories of cultivated plants inevitably miss some taxa, and location-based inventories are inherently limited in the amount of cultivated plant diversity they can capture. Cumulative taxon richness across lists reached 78.4% of the estimated maximum richness (Figure 2e, Table S1). This indicates that although taxon accumulation is plateauing as additional lists are added to WorldCuP (i.e., the rate of accumulation is decreasing), the cumulative number of taxa will most likely continue to increase. These patterns varied considerably across regions. Europe and Asia both showed relatively steep initial increases in taxon richness, reaching 79.0% and 64.3% of estimated maximum richness, respectively. In contrast, taxon accumulation in the Americas was more gradual with a longer tail, reaching 63.2% of estimated maximum richness. Estimated maximum richness in Africa and Oceania had extremely broad confidence intervals as a result of the relatively sparse and/or heterogeneous data in these regions (Table S1); for example, Oceania's steep increase and lack of a plateau can be explained by the inclusion of large national-scale cultivated floras in Australia and New Zealand along with a relative lack of other sources.

Because the spatial extent of WorldCuP regions is highly variable and the regions can overlap, we anticipate two distinct use cases for the data. First, WorldCuP allows for in-depth regional studies of cultivated plants. Several studies using WorldCuP have already been conducted for the regional cultivated floras of the United States (Kinlock

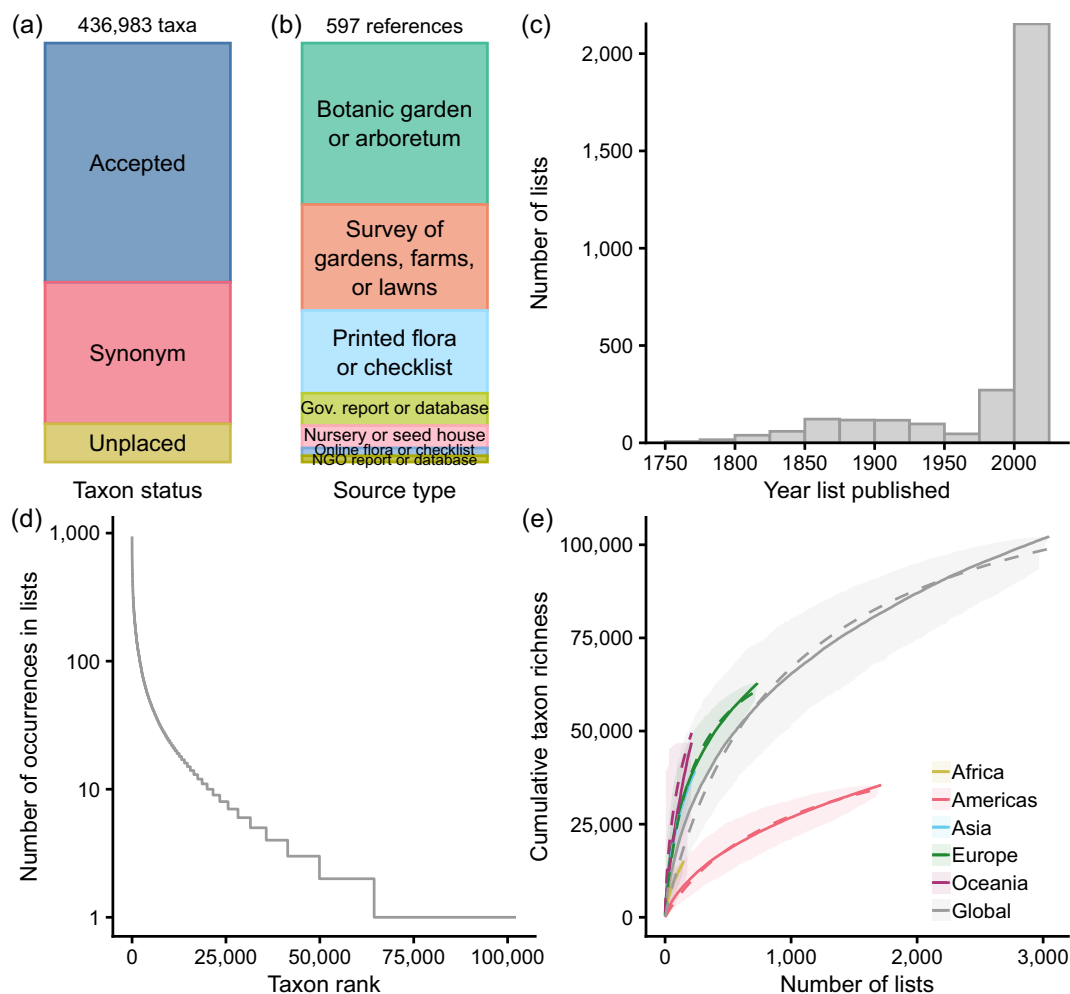
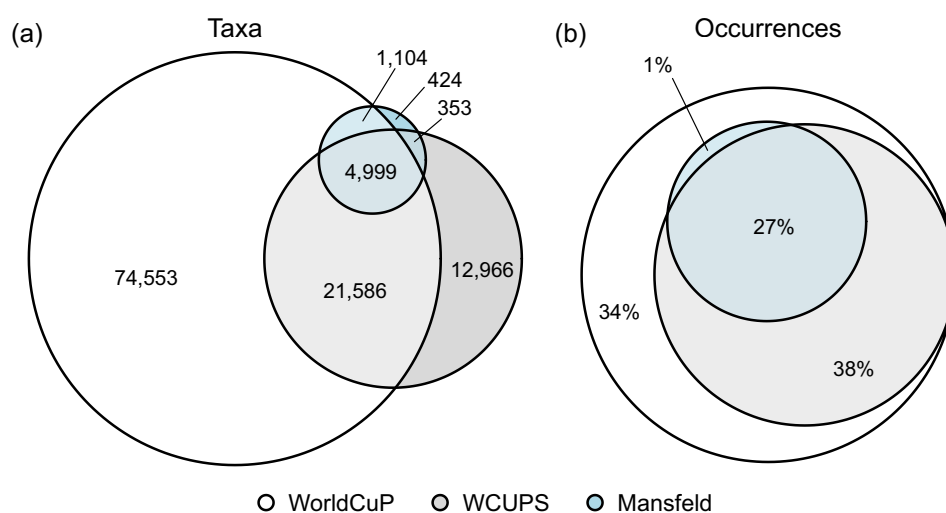


FIGURE 2 Characteristics of the WorldCuP database. (a) Taxonomic statuses for all cultivated plant taxa (including species, infraspecific taxa, and cultivars) included in WorldCuP, according to the WCVF taxonomic backbone. Cultivar names were not included when standardizing taxon names. (b) Source types for all references in WorldCuP. (c) Frequency of WorldCuP lists according to the year each list was published. Seven lists published between 1375 and 1749 are not shown. (d) Rank-occurrence curve showing the number of occurrences in WorldCuP lists (log-scaled) for each cultivated plant taxon, ranked from most to least frequently occurring. (e) Species accumulation curve showing cumulative accepted taxon richness of WorldCuP lists. Curves are shown for the entire WorldCuP dataset ("Global") and for each global realm. Solid lines and bands show the mean $\pm$ 95% quantile-type confidence intervals across 1000 permutations. Dashed lines show estimated fits to Clench's equation.

FIGURE 3 (a) Euler diagram showing overlaps among the World Cultivated Plants (WorldCuP) database, the World Checklist of Useful Plant Species (WCUPS) database, and Mansfeld's World Database of Agricultural and Horticultural Crops. Only accepted taxa according to the WCVF backbone are included. (b) Euler diagram showing cultivated plant taxon occurrences in lists in WorldCuP and the percentage of these occurrences that are in WCUPS and/or in Mansfeld's database.



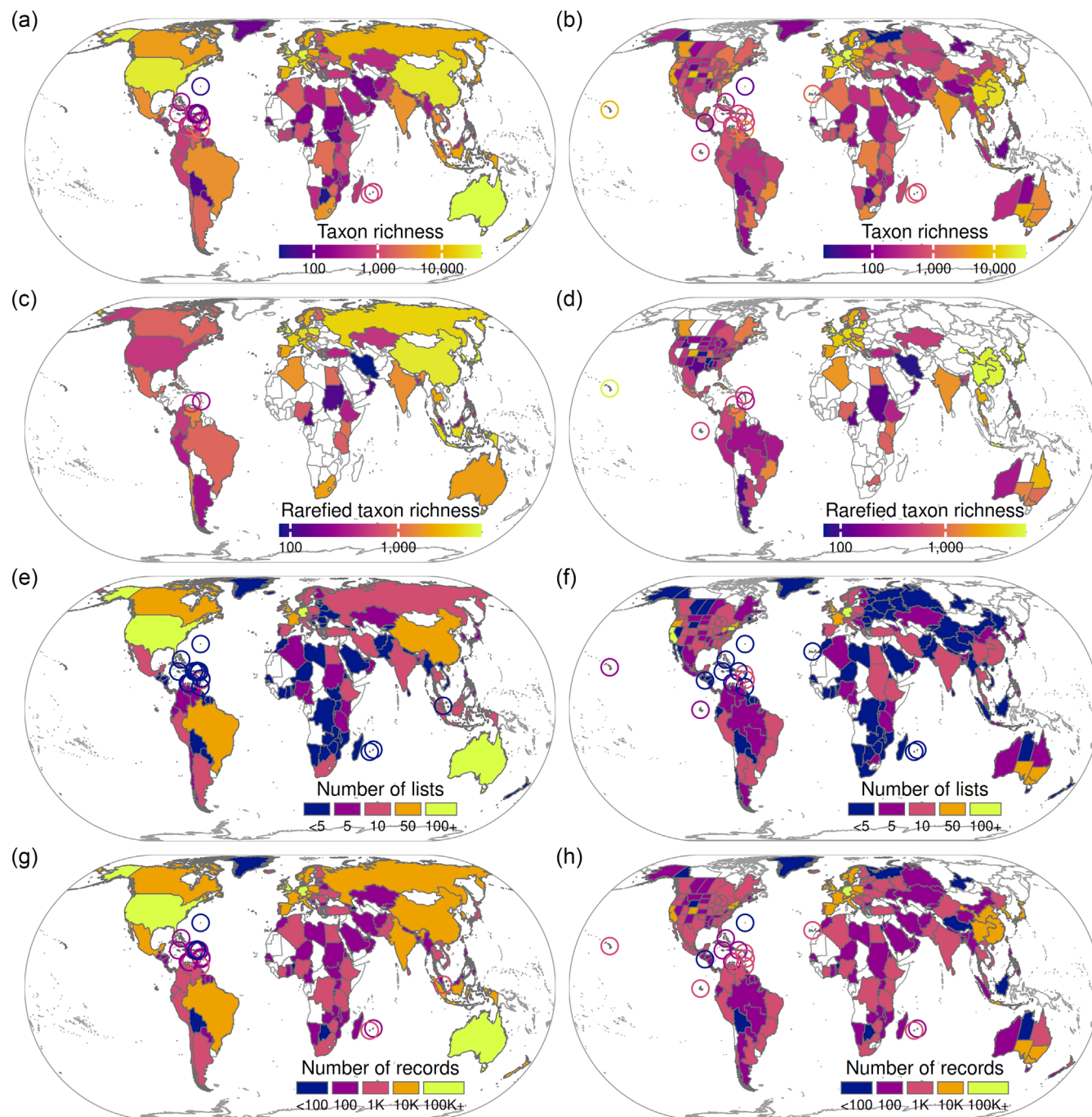


FIGURE 4 Maps of the World Cultivated Plants (WorldCuP) database resolved to International Organization for Standardization (ISO) 3166-1 countries (left column) and Taxonomic Databases Working Group (TDWG) level-3 regions (right column). WorldCuP regions larger than ISO countries or TDWG level-3 regions were removed before resolving the data. (a,b) Raw and (c,d) rarefied cultivated plant taxon richness in regions. Rarefaction involved subsampling each region to $n = 5$ lists and calculating the mean after 1000 iterations. Number of unique (e,f) lists and (g,h) records in WorldCuP in each region. Only taxa that matched accepted names in the WCVF taxonomic backbone were included. Richness is on the base-10 log scale. Islands or island groups are indicated with circles. The Eckert IV projection is used.

et al., 2022; Kinlock et al., 2025), China (Dong et al., 2024, 2025), and Southern Africa (Omer et al., 2021). Second, WorldCuP allows for global-scale analyses of cultivated species occurrences resolved to regions with coarse spatial grain. Although an in-depth analysis of cultivated plant diversity across global regions lies outside the scope of

the current study, we will demonstrate how the WorldCuP database can be used for these types of analyses. Importantly, when resolving occurrences to regions with coarse spatial grain, occurrences from WorldCuP regions that are larger than the regions of interest should be removed.

In the current study, we present the WorldCuP database resolved to countries and TDWG level-3 regions. These regions represent primarily political rather than biogeographical borders. However, political borders are relevant for cultivated species, as they reflect the culture of the people living there, which in turn influences the plants they choose to cultivate, and regulations influencing plant introduction, movement, and sale act within political borders. We found that cultivated plant taxon richness was highest in Australasia, the United States, Western Europe, and China and was comparatively low across much of Africa, southwest Asia, and parts of Central and South America (Figure 4a). Observing richness across TDWG level-3 regions, which are generally of finer grain, revealed that cultivated plant richness was highest in eastern North America, the Pacific coast of North America, southern Brazil, eastern China, eastern Australia, and the North Island of New Zealand (Figure 4b).

The same regions with high cultivated plant taxon richness in WorldCuP also have well-established horticultural industries and botanical institutions (Dehnen-Schmutz et al., 2010; McCracken, 1997), consistent with the patterns that we found. At the same time, the spatial distribution of WorldCuP data was unequal across global regions (Figure 4e–h), and regions with more lists tended to have more taxa (Pearson's r between richness and number of lists = 0.49 in countries and 0.59 in TDWG regions). This suggests that the patterns in richness that we observed reflect a combination of true regional variation in cultivated plant diversity and spatial biases in WorldCuP.

To quantify the influence of spatial biases in the distribution of WorldCuP data, we estimated rarefied taxon richness. Typical ecological rarefaction analyses use rarefied taxon richness as an estimate of true species richness, which assumes that sampling effort is equal across sampling units. This assumption does not hold for WorldCuP, where lists differ greatly in spatial resolution and effort (e.g., a national-scale cultivated flora versus a survey of plants cultivated in domestic gardens in a single village). We therefore consider rarefaction of WorldCuP data not as an estimate of true species richness, but as an assessment of the heterogeneity of lists across regions. Raw taxon richness was correlated with rarefied taxon richness (Pearson's r = 0.59 in countries and 0.76 in TDWG regions). Regions with relatively high rarefied taxon richness (e.g., in Europe and East Asia, Figures 4c,d) tended to have many taxa with divergent taxonomic composition, while regions with lower rarefied taxon richness (e.g., in the Americas) tended to include fewer taxa with more overlapping composition. Lower rarefied richness in the Americas is consistent with the shallower species accumulation curve we observed in this region (fewer new taxa accumulate with increasing sampling effort, Figure 2e). This may also reflect differences in the types of lists available in each region: The Americas have a higher density of nursery/seed house catalogs and surveys of domestic gardens or farms (Figure S2), which may contain more similar sets of ornamental and crop species. In comparison, Europe has denser botanic garden and arboretum records (Figure S2), which may contain more rarely cultivated taxa, possibly maintained solely for scientific or conservation value. We recommend that users of the WorldCuP database estimate rarefied diversity metrics, subsampling either lists or records within

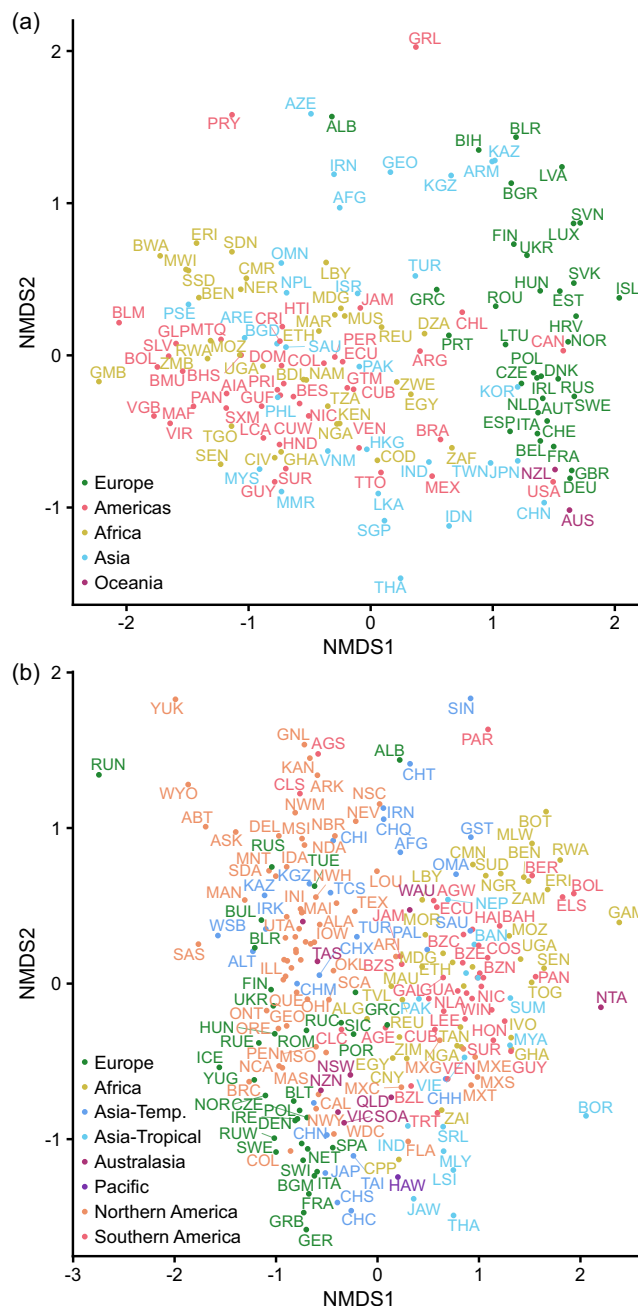


FIGURE 5 (a) Nonmetric multidimensional scaling (NMDS) ordination of cultivated plant taxon composition across International Organization for Standardization (ISO) countries (stress = 0.102). Points are labeled with ISO alpha-3 country codes and colored by global realms. (b) NMDS ordination of composition across Taxonomic Databases Working Group (TDWG) level-3 regions (stress = 0.111). Points are labeled with TDWG level-3 codes and colored by TDWG level-1 regions. Only the first two out of the three ordination dimensions are shown.

regions to a comparable number, in order to quantify these spatial biases. Users may also want to consider restricting analyses to regions with more than some minimum number of lists and/or records.

In an NMDS ordination, the cultivated flora of European countries roughly clustered together, though Canada, the United States,

Australia, New Zealand, China, Japan, and South Korea had similar cultivated taxon composition to European countries (Figures 5a, S3). Cultivated plant taxon composition also appeared to be structured by latitude; high-latitude regions tended to have high values on the NMDS1 and NMDS2 axes in the ISO country ordination. Latitude, as a main driver of climatic zones and native species diversity (Hillebrand, 2004; Kinlock et al., 2018), likely determines climate suitability for cultivated plants. An ordination of TDWG level-3 regions showed considerable variation in composition across North America, China, and Australasia (Figures 5b, S3). The cultivated floras of South America, Central America, the Caribbean, and Africa roughly overlapped in taxon composition. This may reflect shared environmental constraints for cultivated plants in (sub)tropical or arid climates, the legacy of the Atlantic slave trade and the plantation complex (Carney & Rashford, 2013; Schiebinger & Swan, 2005), the emergence of several African and South American countries as major production regions for the horticultural industry (Hinsley et al., 2025), and/or shared cultural preferences among African diaspora communities (Voeks, 2009). Asian regions occupied the broadest range of ordination space. In addition to being the largest landmass, Asia was also a very early center of plant domestication (Larson et al., 2014), meaning that Asian cultivated floras have had significant space and time to diverge.

The WorldCuP database is publicly available as a set of R data frames stored using the compressed R Data Serialization (RDS) format. For users who do not use the R programming language, the same set of tables is available as a set of CSV files compressed as a ZIP file. We also provide an R script that demonstrates how to work with the WorldCuP database using the “data.table” R package (Barrett et al., 2025) or “tidyverse” R tools (Wickham et al., 2019). Version 1.0 of the WorldCuP database, including all tables, metadata, and guides, is currently available on Zenodo. Future updates to the WorldCuP database will be versioned in the same Zenodo repository in alignment with FAIR (Findable, Accessible, Interoperable, and Reusable) data principles (Wilkinson et al., 2016). More information about the WorldCuP database is detailed in the “Guide to the WorldCuP database” PDF file also available in the Zenodo repository. Users of WorldCuP data are encouraged to contact any of the authors regarding potential errors in the data or to contribute region-specific cultivated plant lists.

AUTHOR CONTRIBUTIONS

Nicole L. Kinlock and Mark van Kleunen designed the research; Nicole L. Kinlock performed the research with input from Mark van Kleunen and Amy J. S. Davis; Nicole L. Kinlock collected and interpreted the data with input from Mark van Kleunen; Nicole L. Kinlock wrote the first draft of the manuscript; and all authors edited the manuscript.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

All data, metadata, guides, and code that are a part of the World Cultivated Plants (WorldCuP) database are publicly available on Zenodo at <https://doi.org/10.5281/zenodo.18323695>. Data from previously published sources are individually attributed to their original authors in the Supporting Information. Their inclusion here does not affect authorship of, or recognition for, the original work, which remains fully credited to the cited sources.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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